



## INFORMATION RELEASE MEMO

# Fluids

## CAT<sup>®</sup> PACKAGED FLUIDS SHELF LIFE AND STORAGE GUIDELINES



Caterpillar has developed specific guidelines to achieve the maximum shelf life for lubricating oils including all viscosity grades of Cat<sup>®</sup> DEO, Cat DEO-ULS<sup>™</sup>, Cat DEO SYN<sup>™</sup>, Cat DEO-ULS SYN, Cat DEO Cold Weather, Cat MAEO, Cat NGE<sup>™</sup>, Cat NGE EL250, Cat NGE EL350, Cat NGE Advanced 40, Cat NGE Ultra 40, Cat HYDO<sup>™</sup> Advanced, Cat TDTO<sup>™</sup>, Cat TDTO-TMS<sup>™</sup>, Cat TDTO Cold Weather 0W-20, Cat FDAO<sup>™</sup>, Cat FDAO Syn, Cat GO, Cat GO SYN, Cat MTO, and Cat Compactor Oil are covered by these guidelines.

These guidelines are also applicable for Cat coolants: Cat ELC, Cat ELI, Cat DEAC<sup>™</sup>, Cat NGEC<sup>™</sup>, and Cat SCA. These guidelines are also applicable for all NLGI grades of Cat greases: Cat Arctic Platinum, Cat Desert Gold, Cat Ultra 5Moly, Cat Advanced 3Moly, Cat Multipurpose Grease, Multipurpose Marine Grease, Cat High Speed Ball Bearing Grease, Cat High Vacuum Grease, Cat Ultra Synthetic Generator Grease, Cat Hammer Paste, and Cat White Assembly Grease.

**Shelf Life**

The shelf life of a product begins from the date of packaging and is contingent upon the location and temperature guidelines.

| Product                                        | Shelf Life (months) | Temperature Range |
|------------------------------------------------|---------------------|-------------------|
| Cat NGE0 Products <sup>[1]</sup>               | 36                  | Moderate Temps    |
| Cat Bio Hydo Advanced <sup>[1]</sup>           | 24                  | Moderate Temps    |
| All other Cat Oils <sup>[1]</sup>              | 60                  | Moderate Temps    |
| ELC (Premix & Concentrate) <sup>[2], [3]</sup> | 96                  | >-20°C <35°C      |
| ELI (concentrate) <sup>[2], [3], [4]</sup>     | 12                  | >0°C <35°C        |
| Extender                                       | 12                  | >0°C <35°C        |
| DEAC (concentrate) <sup>[2], [3]</sup>         | 18                  | >-20°C <35°C      |
| DEAC & NGEC (premix 50/50) <sup>[2], [3]</sup> | 18                  | >-20°C <35°C      |
| SCA (Supplemental Coolant Additive)            | 12                  | >0°C <35°C        |
| Grease                                         | 60                  | Moderate Temps    |

<sup>[1]</sup> Shelf is based on indoor storage. If stored outdoors, shelf life is half the stated time (i.e. 60 month shelf life becomes 30 months)

<sup>[2]</sup> It is strongly advised that coolant in translucent packages (such as 275 gal/1000L plastic totes) should not be exposed to direct sunlight as this can degrade the color dyes and can result in fading or discoloration of the color over time. This reaction can be accelerated if coupled with high ambient temperature. It is therefore advisable to store coolant with translucent packaging indoors to avoid this issue.

<sup>[3]</sup> Never recycle containers, always use new. As with any antifreeze coolant, the use of galvanized steel is not recommended for pipes or any other part of the storage/mixing installation.

<sup>[4]</sup> ELI is water based and will freeze - it is recommended that this product be stored where it will not be subjected to temperatures below 0°C

If shelf life of the product is exceeded, regardless of storage location, the following action should be taken based on fluid type:

**Cat Greases**

No direct evaluation criteria exist. Small quantities should be disposed of per appropriate local regulations. If a large quantity of product exists which has passed the storage shelf life guidelines, contact Caterpillar via a Dealer Service Network (DSN) request to get to the appropriate Caterpillar contact in order to determine any alternative course of action.

**Cat DEAC, NGEAC, & ELC Products**

Obtain a 4-ounce (125-mL) sample from the bottom of the storage container. If visible contamination, likely from sediment formation or precipitated additive, is present, the product should not be used. Dispose of the product per appropriate local regulations. Otherwise, submit the sample with proper product identification for S·O·S<sup>SM</sup> Level 2 coolant analysis. If S·O·S results are "green," indicating parameters such as pH, additive levels, and glycol oxidation products are within typical levels for a new coolant sample, the coolant shelf life can be extended by 6 months. If the coolant from the tested container has not been consumed within this 6 month period, repeat this process to determine suitability for additional extension.

**Cat ELI, SCA, & ELC Extender**

Obtain a small sample from the bottom of the storage container. If visible contamination is present, likely from precipitated additive, the product should not be used. Dispose of the product per appropriate local regulations. Otherwise, samples of at least 4 oz. (125-mL) of these products with proper product identification following dilution with water can be sent for S·O·S Level 2 coolant analysis. Samples of the undiluted solutions should NOT be sent. Dilute each with water of acceptable quality (per water quality requirements specified in SEBU6250) according to the following target percentages:

Cat ELI: 7.5% by volume

Cat SCA: 7% by volume

Cat ELC Extender: 2% by volume

If Level 2 S·O·S coolant analysis results are "green," indicating parameters such as pH and additive levels are within typical levels for a new sample, the shelf life of these products can be extended by 3 months. If the product from the tested container has not been consumed within this 3 month period, repeat this process to determine suitability for additional extension.

**Cat Oil Products**

Obtain a 4-oz (125-mL) sample from the bottom of the storage container. If visible contamination is present, likely from sediment formation or additive precipitation, or if separated water droplets or layer is visible, the product should not be used. Dispose of the product per appropriate local regulations. Otherwise, submit the sample with proper product identification for S·O·S oil analysis. If S·O·S analysis results are "green," indicating parameters such as oxidation, additive levels, and viscosity are within typical levels for a new oil sample, the oil shelf life can be extended by 6 months. If the oil from the tested container has not been consumed within this 6 month period, repeat this process to determine suitability for additional extension.

### **Indoor Storage Guidelines**

Ideally lubricants should be stored indoors; in moderate temperatures; and located away from possible sources of contamination. The tops should always be maintained free of any dust, dirt, moisture, or debris.

- Drums should be stored upright. It is recommended that the 1U-6156 covers be utilized. If covers are unavailable, the tops should be kept clear of any debris collection.



- If drums are received from the supplier horizontally, they should be placed on proper storage racks with the bungs positioned at 9 and 3 o'clock. This will ensure that they are covered by the drum contents, thus minimizing moisture migration and drying of the seals



- Smaller package quantities should also be stored indoors in locations that are free of airborne contaminants and away from heaters, steam lines, or other heat generating equipment. Covers can also be utilized to eliminate the possibility of debris collection on the tops.





### **Outdoor Storage Guidelines**

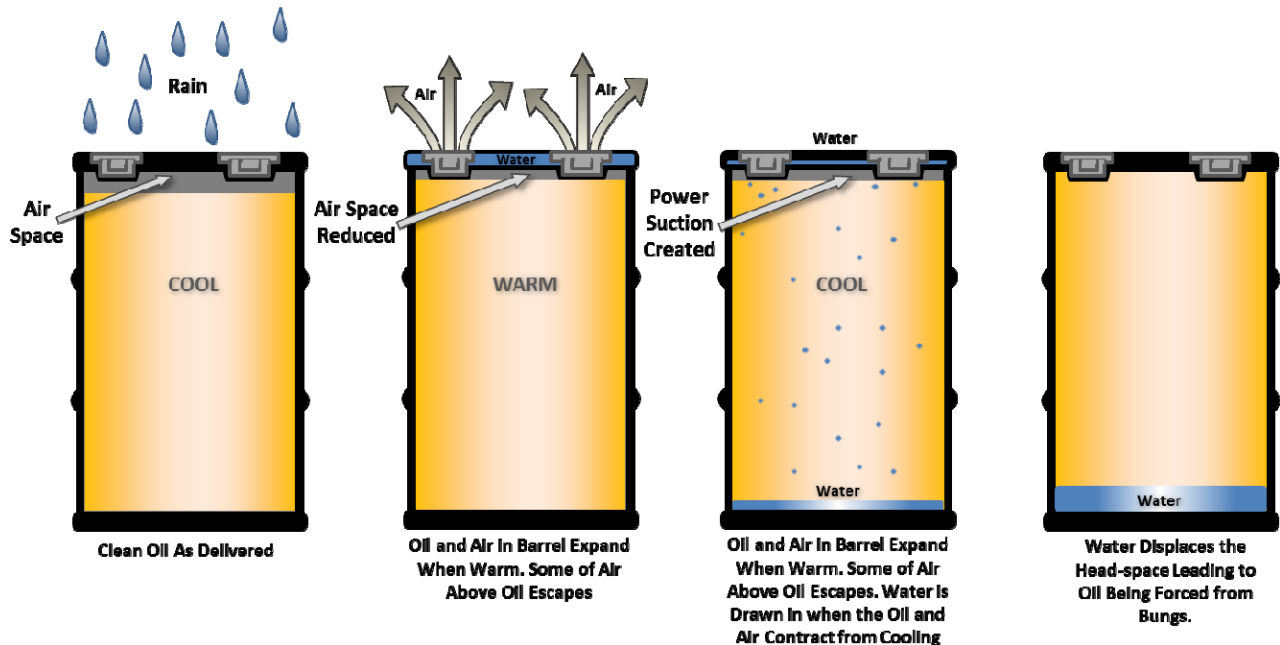
Outdoor storage should be avoided if possible. Besides destroying container labels, widely varying temperatures will cause expansion and contraction of container seams and may lead to leakage. Water can be sucked past the seals as the drum and its contents expand and contract, destroying the oil's performance capabilities. When containers are stored outdoors there are two primary methods for drum storage:

- The preferred method of drum storage is upright with protection from the elements using tight fitting plastic covers like the Cat 1U-6156. This reusable cover can be snapped onto the top of the drum, thus keeping the drum top clean and its contents ready for use.



- If no covers are available, drums should be covered in plastic sheeting if upright or should be placed on their sides with the bungs kept tight and on proper storage racks.
- Position the drums so that the bungs are at 9 and 3 o'clock. This will ensure that the bungs are covered by drum contents thus minimizing moisture migration and drying of seals.

- Drums should never be stored upright without weather protection as water and debris can collect around the bungs. Constant changes in ambient temperature can cause variations in internal pressure which in turn leads to “breathing” causing the water to be drawn into the drum through the bungs.



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 For access and tutorial information: <https://gdsn.cat.com/mscrm>

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